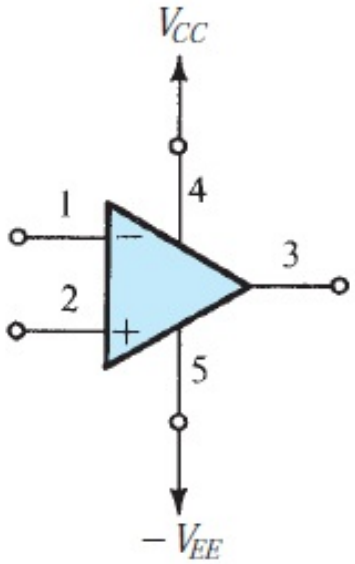
Operational Amplifiers (OPAMPs)

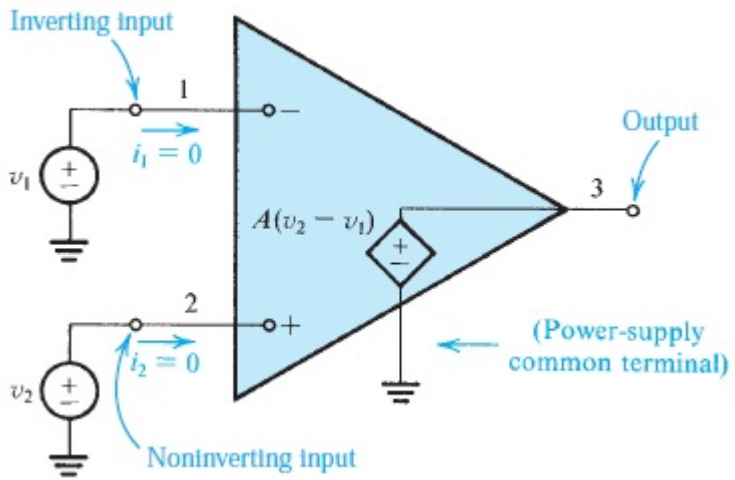
- From a signal point of view the op amp has three terminals: two input terminals and one output terminal. The circuit symbol of an OPAMP is shown in the figure below.



- Terminals 1 and 2 are input terminals, and terminal 3 is the output terminal. Note that amplifiers require DC power to operate. Most IC OPAMPS require two DC power supplies. Two terminals, 4 and 5, are brought out of the op-amp package and connected to a positive voltage V_{CC} and a negative voltage $-V_{EE}$, respectively.

The Ideal OPAMP

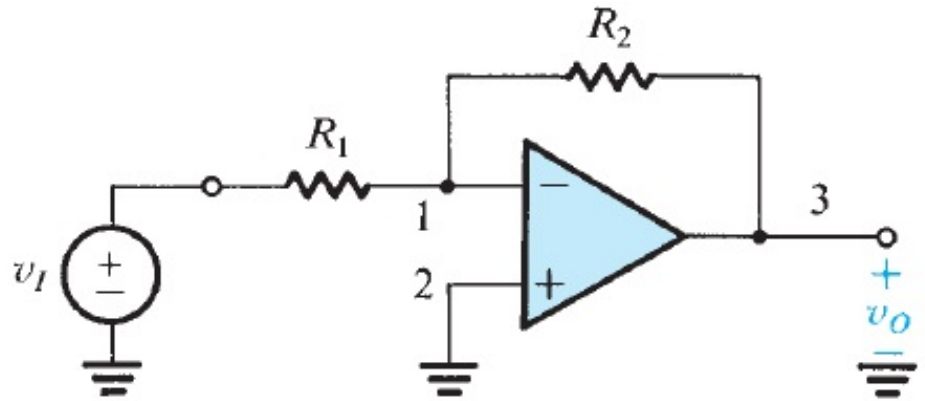
- In addition to the three signal terminals and the two power-supply terminals, an OPAMP may have other terminals for specific purposes. These other terminals can include terminals for frequency compensation and terminals for offset nulling.
- The OPAMP is designed to sense the difference between the voltage signals applied at its two input terminals (i.e., the quantity $v_2 - v_1$), multiply this by a number A, and cause the resulting voltage $A(v_2 - v_1)$ to appear at output terminal 3. Thus $v_3 = A(v_2 - v_1)$.
- Note that when we talk about the voltage at a terminal we mean the voltage between the terminal and ground. Therefore, v_1 means the voltage applied between terminal 1 and ground.



The Ideal OPAMP

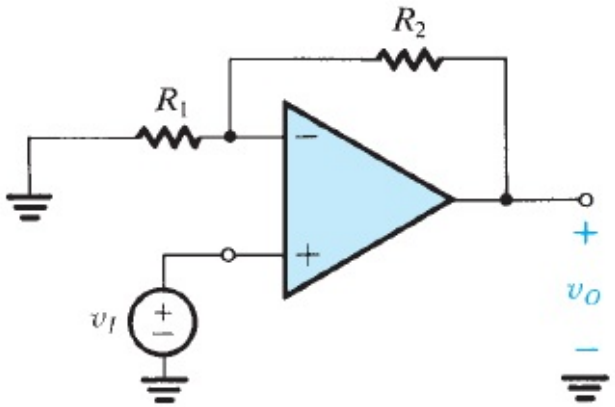
- OPAMPS are not used alone; rather, the op amp is connected to passive components in a feedback circuit. There are two such basic circuit configurations employing an OPAMP and two resistors: the inverting configuration, and the noninverting configuration. In both configurations, the following hold:
  i) There is no current flow into the input terminals of OPAMPs.
  ii) The potential difference between the input terminals of an OPAMP is zero.

The Inverting Configuration



- The inverting configuration consists of one OPAMP and two resistors R_1 and R_2. R_2 is connected from the output terminal of the OPAMP, terminal 3, back to the inverting or negative input terminal, terminal 1. We speak of R_2 as applying negative feedback; if R_2 were connected between terminals 3 and 2 we would have called this positive feedback. Note also that R_2 closes the loop around the OPAMP. In addition to adding R_2, we have grounded terminal 2 and connected R_1 between terminal 1 and an input signal source with a voltage v_i. The output of the overall circuit is taken at terminal 3.
- The output voltage can be calculated as $-\frac{R_2}{R_1}v_i$, i.e., the input voltage is inverted by a factor of R_2/R_1.

The Noninverting Configuration

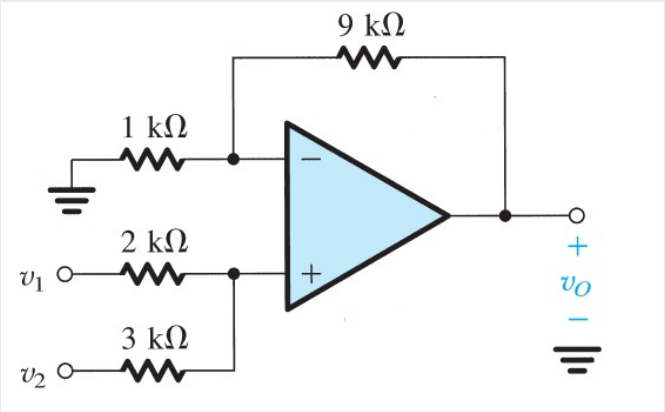


- In the noninverting configuration, the input signal v_i is applied directly to the positive input terminal of the opamp while one terminal of R_1 is connected to ground.

  The output voltage can be calculated as $v_o = v_i \left(1 + \dfrac{R_2}{R_1}\right)$

Examples

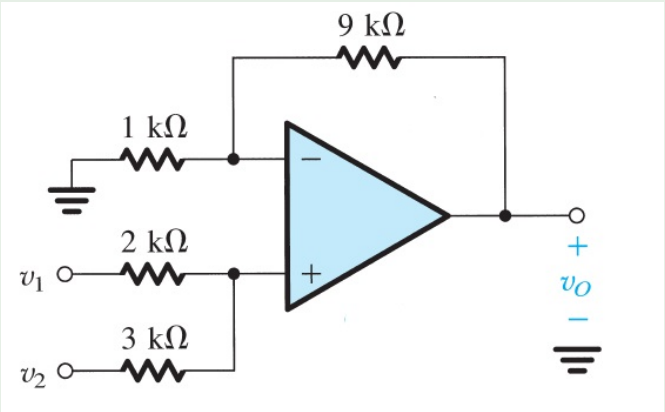
Example



Consider the circuit illustrated above. Find v_o in terms of v_1 and v_2.

Examples

Example

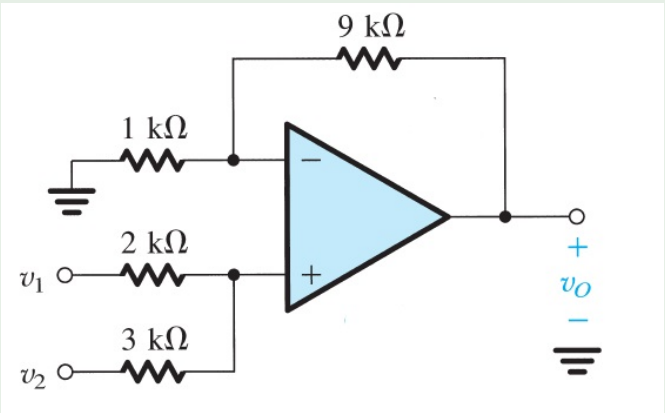


Consider the circuit illustrated above. Find v_o in terms of v_1 and v_2.

Answer: $v_o = 6v_1 + 4v_2$

Examples

Example



Consider the circuit illustrated above. Find v_o in terms of v_1 and v_2.

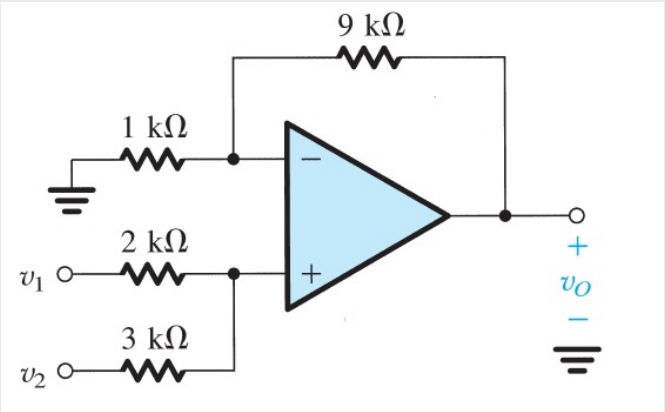
Answer: $v_o = 6v_1 + 4v_2$

Example

If $1k\Omega$ resistor is disconnected from ground and connected to a third signal source v_3, determine v_0 in terms of v_1, v_2, and v_3.

Examples

Example



Consider the circuit illustrated above. Find v_o in terms of v_1 and v_2.

Answer: $v_o = 6v_1 + 4v_2$

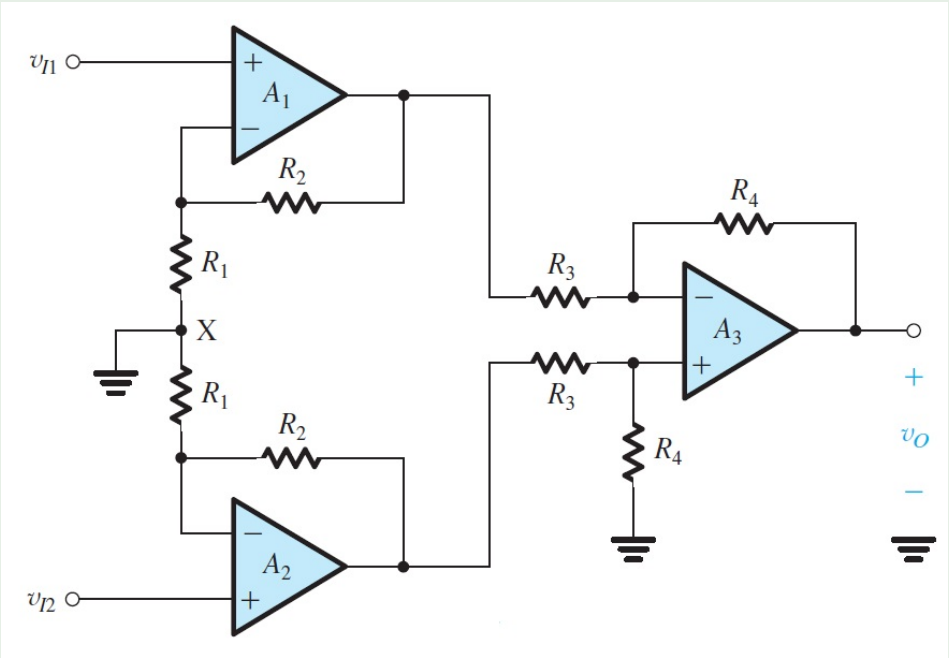
Example

If $1k\Omega$ resistor is disconnected from ground and connected to a third signal source v_3, determine v_0 in terms of v_1, v_2, and v_3.

Answer: $v_o = 6v_1 + 4v_2 - 9v_3$

Examples

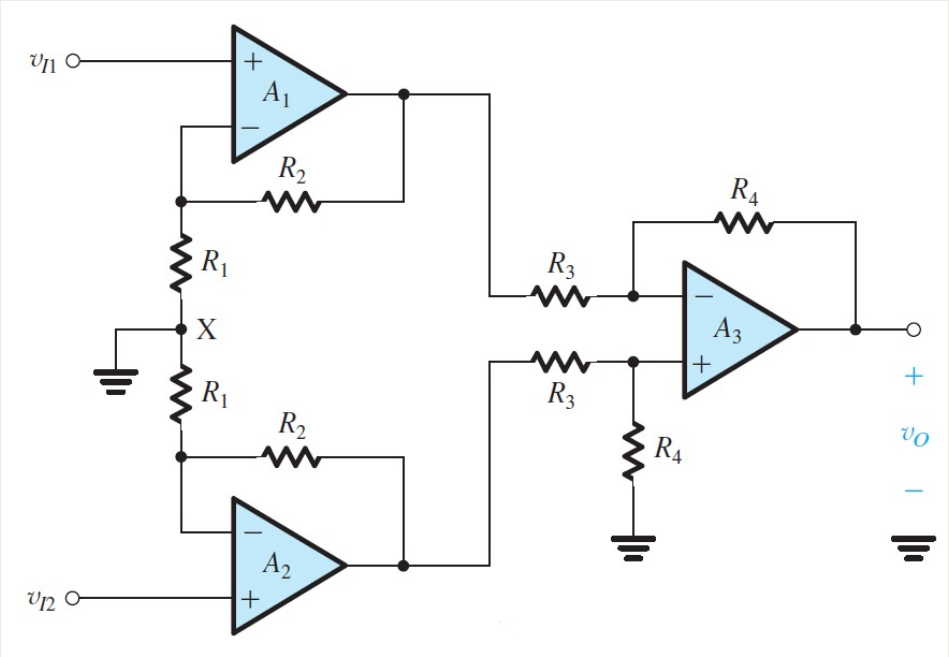
Example



Consider the circuit illustrated above. Find v_o in terms of v_{I1} and v_{I2}.

Examples

Example

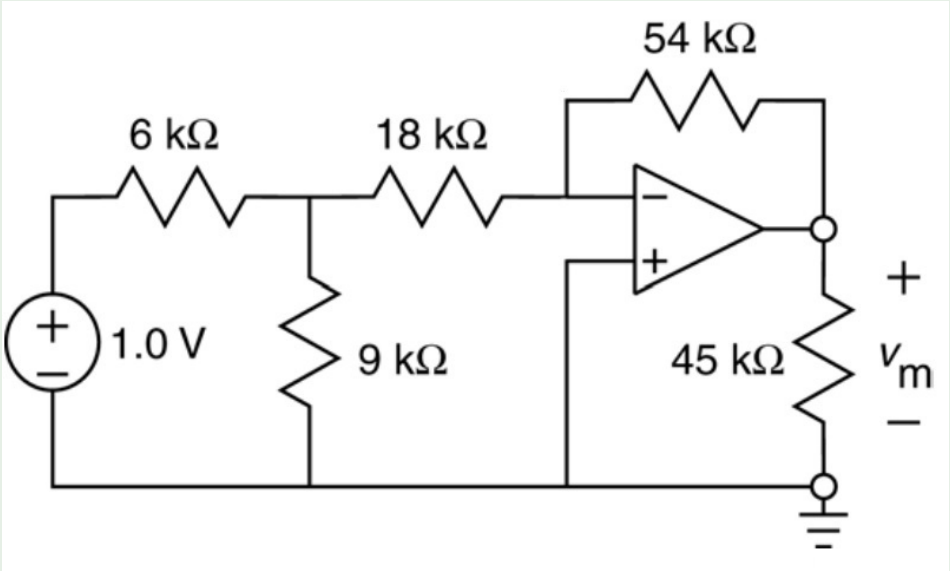


Consider the circuit illustrated above. Find v_o in terms of v_{I1} and v_{I2}.

Answer: $v_o = \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)(v_{I2} - v_{I1})$

Examples

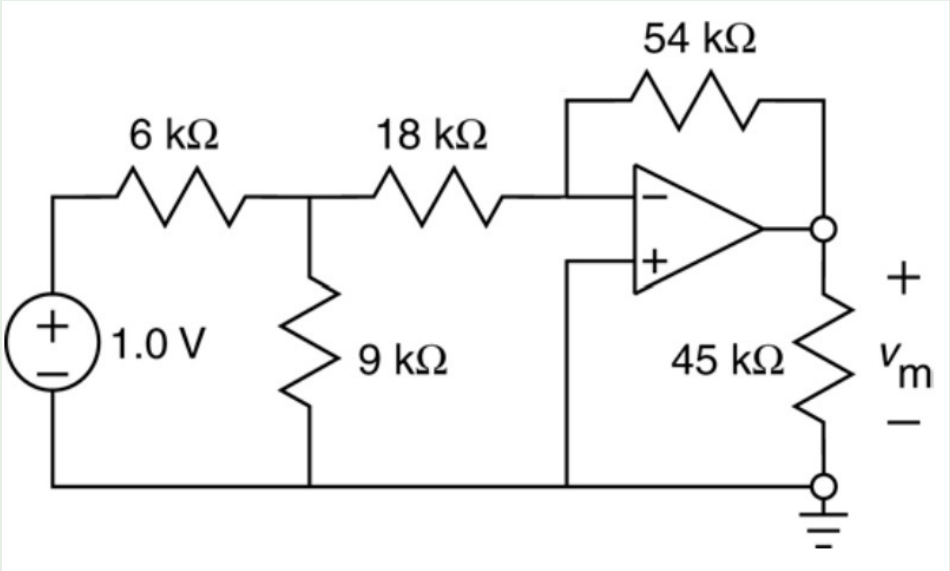
Example



Consider the circuit illustrated above. Find the value of v_m.

Examples

Example



Consider the circuit illustrated above. Find the value of v_m.

Answer: $-1.5V$